

Deep Learning for Solving Dynamic Models

Session 2: Deep Equilibrium Nets, the method and the Brock–Mirman benchmark

Simon Scheidegger

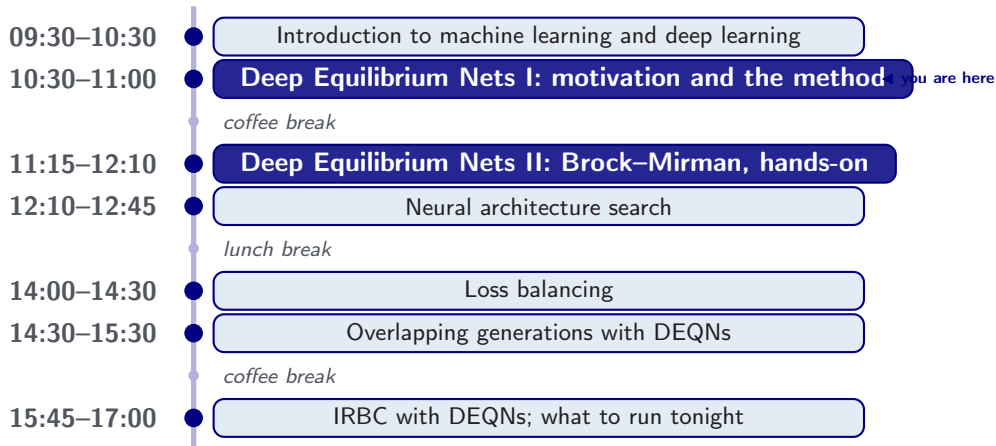
HEC, University of Lausanne

Based on Azinovic, Gaegauf & Scheidegger (2022), *International Economic Review* 63(4),
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Day 1 at a Glance



Roadmap of This Lecture

I. Motivation

- ▶ Heterogeneity, high dimensions, kinks, irregular ergodic sets

II. From neural networks to DEQNs

- ▶ Self-supervised loss, the training algorithm, where DEQNs sit

III. Brock–Mirman benchmark

- ▶ Euler equation, hard constraints, residual and loss
- ▶ Quadrature, diagnostics, validation, loss kernels

IV.–V. Loss balancing, NAS (Session 3)

- ▶ Pathologies (a) and (b) get their own session: 12:10 and 14:00

VI. Further pathologies, HPC, wrap-up

- ▶ Pathologies (c) to (e), data parallelism, hands-on preview

Learning Objectives

By the end of this lecture you should be able to:

- ▶ Formulate dynamic stochastic economic models as equilibrium systems and identify the computational challenges in high dimensions
- ▶ Replace grid-based collocation with **self-supervised** neural-network training that uses equilibrium errors as the loss
- ▶ Build and train Deep Equilibrium Nets (DEQNs), and validate one against the Brock–Mirman closed form
- ▶ Diagnose and cure the two most common failure modes in practice: **loss-scale imbalance** and **poor architecture choice**
- ▶ Recognize the three residual-only pathologies that remain: lazy learning, learning from scratch, and self-confirming coverage
- ▶ Know when to reach for adaptive loss balancing (ReLoBRaLo), automated architecture search, and data parallelism

Part I

Motivation

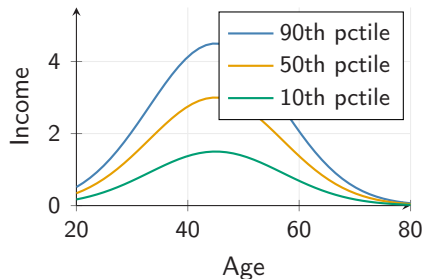
Heterogeneity Matters

The questions: inequality over the life cycle, idiosyncratic versus aggregate risk, distributional effects of policy.

The punchline: **hand-to-mouth** households have a quarterly MPC of roughly 0.3 to 0.5, against about 0.05 for unconstrained ones. A representative agent averages these out and misses the policy response.

Kaplan, Moll & Violante 2018; Bilbiie 2008;
Krueger, Mitman & Perri 2016

⇒ Heterogeneity is one of several forces that **blow up** the state dimension d_x .



Stylized income-by-age profiles

Dynamic Stochastic General Equilibrium Models

Generic formulation: find policy functions $p(\cdot)$ such that

$$\mathbb{E}_t[\mathcal{R}(\mathbf{x}_t, \mathbf{x}_{t+1}, p(\mathbf{x}_t), p(\mathbf{x}_{t+1}))] = 0$$

with transition law $\mathbf{x}_{t+1} = h(\mathbf{x}_t, p(\mathbf{x}_t), \varepsilon_{t+1})$, where $\mathcal{R}$ collects the equilibrium conditions.

Four computational challenges (Azinovic, Gaegauf & Scheidegger, 2022):

1. **Stochasticity**: nontrivial expectations over ε_{t+1} and over the ergodic distribution
2. **High-dimensional** state space: heterogeneity, many assets, and many shocks all push d_x up
3. **Strong nonlinearities and kinks** from occasionally binding constraints and regime switches
4. **Irregular geometry** of the ergodic set: recurrent states are not a hypercube

No classical method addresses all four jointly: sparse grids cover (1)–(3) but become impractical past $d_x \sim 20$; Gaussian processes cover (1), (2) and (4) but cost $\mathcal{O}(n^3)$ to train.

What Is High-Dimensional?

Dimensions d_x	Grid points $n_g^{d_x}$	Runtime
1	10	10 seconds
2	100	2 minutes
5	10^5	1 day
10	10^{10}	317 years
20	10^{20}	~ 3 trillion years

Assumes $n_g = 10$ grid points per dimension and 1 second per evaluation. This is the curse of dimensionality of lecture 1, now with an economic state on the axis.

Three remedies:

1. **Sparse grids** (Brumm & Scheidegger, 2017)
2. **Random grids** (Rust, 1997), for the restricted class of discrete decision processes
3. **Neural networks** (this lecture)

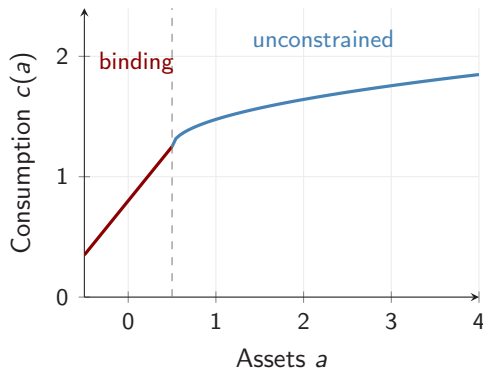
Nonlinearities and Kinks

Where do kinks come from?

- ▶ **Borrowing constraints:** $a_{t+1} \geq \underline{a}$
- ▶ **Zero lower bound:** $i_t \geq 0$
- ▶ **Short-sale and participation:** corner solutions
- ▶ **Default, regime switches, indivisibilities**

Occasionally binding constraints $\Rightarrow$ policy functions are **non-smooth** at the boundary of the binding region.

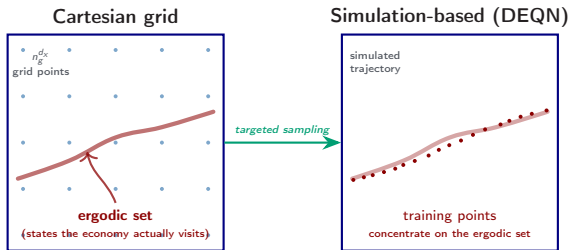
Polynomial and spectral methods lose accuracy; sparse grids can resolve kinks but only for $d_x \lesssim 20$. Neural networks handle kinks natively via ReLU-type activations.



Kinked consumption policy at the borrowing limit

The Ergodic Set Is Not a Hypercube

Grid-based methods tile the full hypercube $\mathbf{x} \in [\mathbf{x}_{\min}, \mathbf{x}_{\max}]^{d_x}$. **But** simulated states drawn from the model's law of motion fill only a **thin, irregular** subset, the **ergodic set** of states the economy actually visits, often $\ll 1\%$ of the hypercube (Judd, Maliar & Maliar, 2011).



A full Cartesian grid **wastes** most of its work in regions the economy never visits, which motivates **simulation-based, grid-free** methods.

What We Need, and What DEQNs Deliver

The wishlist. A numerical method that is

1. **global**: not a local perturbation;
2. content with a small residual error;
3. **grid-free**: avoiding growth like $n_g^{d_x}$;
4. comfortable with kinks and irregular ergodic sets;
5. **parallel-friendly**.

Three ingredients (Azinovic, Gaegauf & Scheidegger, 2022)

1. **Equilibrium error as the loss**: use $\mathcal{R}(\cdot) = 0$ directly, with **no labeled data**
2. **Stochastic gradient descent** on the parameters θ
3. **Simulated paths**: draw training points from the model's own **ergodic distribution**

⇒ This turns **solving** an economic model into **training** a neural network, while **mitigating** rather than eliminating high-dimensional costs: expectation accuracy, sampling quality, and optimization all still matter.

Part II

From Neural Networks to DEQNs

What We Carry Over From Lecture 1

A deep network composes affine maps with activations, with $\mathbf{a}^{(0)} = \mathbf{x}$:

$$\mathbf{a}^{(\ell)} = g^{(\ell)}(\mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}), \quad \ell = 1, \dots, L-1, \quad \mathcal{N}_{\theta}(\mathbf{x}) = \mathbf{W}^{(L)}\mathbf{a}^{(L-1)} + \mathbf{b}^{(L)}.$$

Three facts we use:

- **Universal approximation**: for any continuous f on a compact set and any tolerance, *some* width suffices, though it may be large (Cybenko 1989; Hornik et al. 1989; Leshno et al. 1993).
- **Backprop is reverse-mode autodiff**: we get ∇_{θ} of *any* scalar built from θ .
- **The output layer is a modeling choice**: softplus gives positivity, sigmoid a rate.

Why this is the whole bridge

Lecture 1 ended by observing that autodiff differentiates any scalar built from θ . So we may put an **Euler equation residual** in the loss and still get ∇_{θ} for free. That is exactly what DEQNs do.

Activations. ReLU handles **kinks**; tanh and Swish are **smooth**, which matters when derivatives of the network enter the loss.

Notation for the Rest of the Lecture

Symbol	Meaning
$\mathbf{x}_t$	state vector, dimension d_x
ε_{t+1}	shock, dimension d_ε
$p(\cdot)$	the true policy function
$\mathcal{N}_\theta$	its network approximation
θ	network weights and biases
$\mathcal{R}$	equilibrium residual

Symbol	Meaning
$J(\theta)$	total training loss
$\mathcal{L}_k$	loss component, $k = 1, \dots, M$
ω_k	weight on component k
N	number of sampled states
Q, w_q	quadrature nodes and weights
η	learning rate

Economic parameters keep their usual names: α capital share, β discount factor, δ depreciation, ϱ persistence, σ_z shock volatility. Balancing and search hyperparameters carry subscripts (α_{1b} , ρ_{1b} , γ) so that they never collide with them.

Supervised Learning and the “Data Problem” in Economics

The standard supervised recipe needs labeled data $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ and minimizes

$$J^{\text{sup}}(\theta) = \frac{1}{N} \sum_{i=1}^N \|y_i - \mathcal{N}_{\theta}(\mathbf{x}_i)\|^2,$$

by stochastic gradient descent, exactly as in lecture 1.

But in computational economics:

- ▶ We solve **functional equations**, not classification or regression tasks
- ▶ There are **no labeled pairs** $(\mathbf{x}_i, y_i)$: nobody hands us the true policy

What we can exploit

We **know** the structural equations $\mathcal{R}(\cdot) = 0$, and we can **check** how well a candidate policy satisfies them.

⇒ Use the **equilibrium error** as the loss function.

Traditional Approach: Time Iteration on a Grid

Algorithm: grid-based collocation

```
Input: grid  $\mathcal{G} = \{\mathbf{x}_1, \dots, \mathbf{x}_{M_g}\} \subset \mathbb{R}^{d_x}$ , initial guess  $p^{(0)}$   
for  $k = 0, 1, 2, \dots$  until convergence do  
  for each grid point  $\mathbf{x}_i \in \mathcal{G}$  do  
    Solve  $\mathcal{R}(\mathbf{x}_i, p^{(k)}(\mathbf{x}_i), p^{(k)}) = 0$  for  $p^{(k+1)}(\mathbf{x}_i)$   
  end for  
  Interpolate  $p^{(k+1)}$  between grid points  
end for
```

Problems:

- ▶ **Cartesian** grids need $M_g = n_g^{d_x}$ points, infeasible beyond $d_x \sim 4-5$ at sensible resolution (10^5 at $d_x=5$, 10^{10} at $d_x=10$)
- ▶ **Sparse grids** (Smolyak, adaptive) extend the practical reach to roughly $d_x \sim 20$
- ▶ Heterogeneous-agent and many-country models often need far more, which is beyond the reach of tensor grids

The DEQN Loss Function

The idea: replace labeled data with **economic equilibrium conditions**. Let $\mathcal{R}(\mathbf{x}, p(\mathbf{x})) = 0$ be the system of equilibrium equations, and define

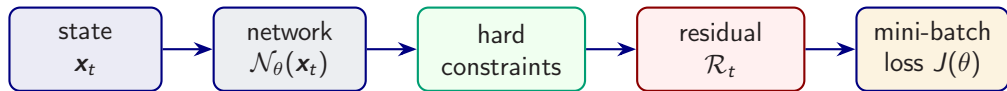
$$J(\theta) = \frac{1}{N} \sum_{i=1}^N \|\mathcal{R}(\mathbf{x}_i, \mathcal{N}_{\theta}(\mathbf{x}_i))\|^2$$

- ▶ $\mathcal{N}_{\theta}(\mathbf{x}_i)$ is the network's **predicted policy** at state $\mathbf{x}_i$, and $\mathcal{R}(\cdot)$ is the **equilibrium error**, for instance an Euler equation residual
- ▶ A small $J(\theta)$ means the policy nearly satisfies the equations on the sampled states
- ▶ **No labels needed**. We call this **self-supervised** learning; it is sometimes labeled “unsupervised” in the narrow no-labels sense

One caveat we will make precise in Part III

When $\mathcal{R}$ contains a conditional expectation, it must be approximated *inside* the norm. Squaring a single simulated draw is a different, and biased, objective.

Inside One DEQN Training Step



- ▶ **Hard constraints** are what the architecture can enforce exactly: budgets, positivity, market clearing.
- ▶ **Soft conditions** are what is left in the loss: Euler equations, KKT conditions, remaining equilibrium conditions.
- ▶ States are drawn from the model's own simulation, so training focuses on the ergodic set.

Training Algorithm for DEQNs

Algorithm 1: Deep Equilibrium Net training

Input: initial state $\mathbf{x}_0$, network $\mathcal{N}_\theta$, learning rate η , episodes E , horizon T_{sim} , steps per episode T_{train}

for episode $j = 1, \dots, E$ **do**

Simulate path: $\mathbf{x}_0 \rightarrow \mathbf{x}_1 \rightarrow \dots \rightarrow \mathbf{x}_{T_{\text{sim}}}$ using $\mathcal{N}_\theta$ and the transition law, then **detach** the states

for gradient step $m = 1, \dots, T_{\text{train}}$ **do**

 Draw mini-batch $\mathcal{B}$ of size B from $\{\mathbf{x}_0, \dots, \mathbf{x}_{T_{\text{sim}}}\}$

$J(\theta) = \frac{1}{B} \sum_{\mathbf{x}_i \in \mathcal{B}} \|\mathcal{R}(\mathbf{x}_i, \mathcal{N}_\theta(\mathbf{x}_i))\|^2$

$\theta \leftarrow \theta - \eta \nabla_\theta J(\theta)$

end for

Stop when the test Euler-error quantiles stop improving over a window of episodes

end for

Output: trained network $\mathcal{N}_{\theta^*}$ approximating the policy function

Note. Training points come from the **model's own simulation**, so the distribution the network is trained on *moves while training*. There is no contraction here and no convergence theorem: the stopping rule has to be empirical.

Where DEQNs Sit in the Literature

Classical global methods:

- ▶ Projection and collocation (Judd 1992)
- ▶ Parameterized expectations (den Haan & Marcet 1990)
- ▶ Krusell & Smith (1998)
- ▶ Adaptive sparse grids (Brumm & Scheidegger 2017)

One sentence for macroeconomists

A DEQN is **parameterized expectations with a neural network**: the same regression on simulated data, with a flexible approximator and SGD in place of a linear projection.

Deep-learning solvers:

- ▶ **DEQNs** (Azinovic, Gaegauf & Scheidegger 2022), this lecture
- ▶ Maliar, Maliar & Winant (2021), a unified framework
- ▶ Duarte (2018), continuous time
- ▶ Fernández-Villaverde et al. (2020), dynamic programming
- ▶ **DeepHAM** (Han, Yang & E 2026), heterogeneous agents
- ▶ Kahou et al. (2021), symmetry and exchangeability

Part III

Brock–Mirman Benchmark

Brock–Mirman (1972): Setup

Why this model. It is the one growth model whose policy function is known in closed form, so every number a DEQN produces can be checked against the truth.

Planner's problem. Choose consumption to maximize expected lifetime log utility,

$$\max_{\{C_t\}_{t=0}^{\infty}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \ln C_t \right], \quad C_t + K_{t+1} = z_t K_t^{\alpha}, \quad K_0 > 0 \text{ given},$$

with full depreciation $\delta = 1$, capital share $\alpha \in (0, 1)$, and discount factor $\beta \in (0, 1)$.

Stochastic productivity. TFP follows an AR(1) in logs, with $|\varrho| < 1$:

$$\ln z_{t+1} = \varrho \ln z_t + \sigma_z \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1).$$

State, control, timing. The state is $\mathbf{x}_t = (K_t, z_t)$ and the control is C_t , with $K_{t+1} = z_t K_t^{\alpha} - C_t$. Both C_t and K_{t+1} are **chosen at date t** , so both are measurable with respect to time- t information. The network takes $(K_t, \ln z_t)$ as input.

That timing assumption is what lets us pull $1/C_t$ out of the expectation two slides from now.

Deriving the Euler Equation (Lagrangian)

Step 1. Attach a multiplier λ_t to each period's budget:

$$\Lambda = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t [\ln C_t - \lambda_t (C_t + K_{t+1} - z_t K_t^\alpha)] \right].$$

Step 2. K_{t+1} appears **twice**, as an outflow at t and through production at $t+1$, so both terms enter. Dividing by β^t ,

$$\frac{\partial \Lambda}{\partial C_t} : \lambda_t = \frac{1}{C_t}, \quad \frac{\partial \Lambda}{\partial K_{t+1}} : \lambda_t = \beta \mathbb{E} [\lambda_{t+1} \alpha z_{t+1} K_{t+1}^{\alpha-1}].$$

Step 3. As λ_t and K_{t+1} are known at t , iterated expectations makes the expectation conditional, and $\lambda_t = 1/C_t$ gives

$$\boxed{\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{\alpha z_{t+1} K_{t+1}^{\alpha-1}}{C_{t+1}} \right]}$$

Marginal utility today equals discounted expected marginal utility tomorrow, times the gross marginal product of capital at $t+1$.

Closed-Form Policy via Guess-and-Verify

Guess. Consumption is a constant fraction of output, $C_t = (1 - s^*) z_t K_t^\alpha$, so the budget gives $K_{t+1} = s^* z_t K_t^\alpha$. Here $s^* \in (0, 1)$ is the **savings rate**, the object the network will learn.

Substitute and solve. With $C_{t+1} = (1 - s^*) z_{t+1} K_{t+1}^\alpha$ the marginal term collapses,

$$\frac{\alpha z_{t+1} K_{t+1}^{\alpha-1}}{(1 - s^*) z_{t+1} K_{t+1}^\alpha} = \frac{\alpha}{(1 - s^*) K_{t+1}}, \quad \text{so} \quad \frac{1}{(1 - s^*) z_t K_t^\alpha} = \frac{\beta \alpha}{(1 - s^*) s^* z_t K_t^\alpha},$$

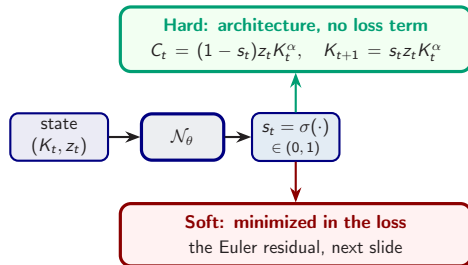
which gives $s^* = \alpha\beta$ and therefore

$$\boxed{C_t = (1 - \alpha\beta) z_t K_t^\alpha, \quad K_{t+1} = s^* z_t K_t^\alpha, \quad s^* = \alpha\beta}$$

Note: z_{t+1} cancels and what survives is *measurable at t* , so $\mathbb{E}_t[\cdot]$ passes straight through. This is special to log utility with Cobb–Douglas production and full depreciation. **Verify:** the Euler equation is necessary, not sufficient, and the candidate also satisfies transversality, since $\beta^t \lambda_t K_{t+1} = \alpha \beta^{t+1} / (1 - \alpha\beta) \rightarrow 0$, which rules out the explosive alternatives.

Brock–Mirman: Hard vs. Soft Constraints

Split the conditions by what the architecture enforces exactly. The network emits a savings rate through a sigmoid head, $s_t = \sigma(\mathcal{N}_\theta(K_t, \ln z_t))$.



The resource constraint is free. The two hard lines add to $C_t + K_{t+1} = z_t K_t^\alpha$, so it holds **identically**, for any θ , at every state.

Why a savings head? A softplus head on C_t would guarantee $C_t > 0$ but could still push $K_{t+1} < 0$. The sigmoid savings head removes *both* failure modes before training starts. With $\delta < 1$, apply the same construction to cash-on-hand.

Brock–Mirman: The Euler Residual

Write the Euler equation as a residual that vanishes exactly at the solution.

Residual in consumption units, the diagnostic

$$e_{\text{REE}} = \frac{\hat{C}_t}{C_t} - 1, \quad \hat{C}_t \equiv (u')^{-1}(\beta \mathbb{E}_t[\cdot]) = (\beta \mathbb{E}_t[\cdot])^{-1},$$

so $e_{\text{REE}} = 10^{-3}$ means consumption is mispriced by **one tenth of one percent**.

Tomorrow's consumption means running the **same network again** at each shock draw:

$$z_{t+1}(\varepsilon) = e^{\varrho \ln z_t + \sigma_z \varepsilon}, \quad s_{t+1} = \sigma(\mathcal{N}_\theta(K_{t+1}, \ln z_{t+1})), \quad C_{t+1} = (1 - s_{t+1})z_{t+1}K_{t+1}^\alpha,$$

with $K_{t+1} = s_t z_t K_t^\alpha$.

Partial depreciation. With $\delta < 1$ the marginal product becomes $\alpha z_{t+1} K_{t+1}^{\alpha-1} + (1 - \delta)$, and $s^* = \alpha\beta$ holds only at $\delta = 1$.

Brock–Mirman: From Residual to Training Loss

Approximate the expectation first, then square. With Q quadrature nodes $\{\varepsilon_q\}$ and weights $\{w_q\}$,

$$\mathcal{R}(\mathbf{x}_i) = 1 - \beta C(\mathbf{x}_i) \sum_{q=1}^Q w_q \frac{\alpha z_{t+1}(\varepsilon_q) K_{t+1}^{\alpha-1}}{C_{t+1}(\varepsilon_q)},$$

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N |\mathcal{R}(\mathbf{x}_i)|^2$$

where i indexes the N simulated states in the batch.

Expectation inside, square outside

Replacing $\mathbb{E}_t[\cdot]$ by a *single* simulated draw and squaring that minimizes

$$\mathbb{E}[e^2] = (\mathbb{E}[e])^2 + \text{Var}(e),$$

whose minimizer trades bias against shock variance. It is *not* the equilibrium policy.

Why Brock–Mirman hides this. Here z_{t+1} cancels, so the residual is zero *pathwise*, realization by realization, and both routes agree.

That cancellation is knife-edge. In any model without it, the single-draw route trains the wrong object while the loss curve still looks healthy.

Gauss–Hermite: Exact in Low Dimensions, Costly in High

Setup. Shock dimension d_ε , with $\varepsilon' \sim \mathcal{N}(0, I_{d_\varepsilon})$: Brock–Mirman has $d_\varepsilon = 1$, an international real business cycle (IRBC) model with n_c countries has $d_\varepsilon = n_c + 1$. Gauss–Hermite uses Q nodes,

$$\mathbb{E}[h(\varepsilon)] = \int_{-\infty}^{\infty} h(\varepsilon) \frac{e^{-\varepsilon^2/2}}{\sqrt{2\pi}} d\varepsilon \approx \sum_{q=1}^Q w_q h(\varepsilon_q),$$

exact for polynomials of degree $\leq 2Q - 1$, spectrally convergent for analytic h .

The rescaling that breaks student code

Library routines such as `numpy.polynomial.hermite.hermgauss` return the *physicists'* rule, for weight e^{-x^2} . Convert with $\varepsilon_q = \sqrt{2}x_q$ and $w_q = \tilde{w}_q/\sqrt{\pi}$. Forgetting this gives a silently wrong expectation.

Tensor product. Combining one-dimensional rules costs Q^{d_ε} nodes per state: for $Q = 3$, $d_\varepsilon = 11$, about 1.8×10^5 .

Stroud-3: Linear Cost in the Shock Dimension

Idea. Keep exactness only to total degree ≤ 3 , with nodes on the principal axes (Stroud 1957, 1971):

$$\mathbb{E}[h(\boldsymbol{\varepsilon}')] \approx \frac{1}{2d_\varepsilon} \sum_{k=1}^{d_\varepsilon} \left[h(+\sqrt{d_\varepsilon} \mathbf{u}_k) + h(-\sqrt{d_\varepsilon} \mathbf{u}_k) \right]$$

so $2d_\varepsilon$ nodes at radius $\sqrt{d_\varepsilon}$, with $\mathbf{u}_k$ the k -th basis vector.

The radius. Matching the second moment,

$\frac{1}{2d_\varepsilon} \cdot 2r^2 = 1$, gives $r = \sqrt{d_\varepsilon}$. **Exact for**
 $\mathbb{E}[\varepsilon'_i] = \mathbb{E}[\varepsilon_i'^3] = 0$, $\mathbb{E}[\varepsilon'_i \varepsilon'_j] = 0$ for $i \neq j$,
 $\mathbb{E}[\varepsilon_i'^2] = 1$.

Where it stops

$\mathbb{E}[\varepsilon_i'^4]$ gives d_ε against a truth of 3: the fourth-moment error **grows linearly in d_ε** , exactly the regime the rule is used for. Check against a finer rule.

Correlated shocks: apply the rule to standard normals, then map through a Cholesky factor.

Which Rule When

Per-state cost: IRBC model with n_c countries, $d_\varepsilon = n_c + 1$ shocks ($Q = 3$ for the tensor product):

n_c	shocks d_ε	tensor product 3^{d_ε}	monomial 2^{d_ε}	speed-up
2	3	27	6	$\sim 5\times$
5	6	729	12	$\sim 60\times$
10	11	177,147	22	$\sim 8,000\times$
20	21	$\sim 10^{10}$	42	$\sim 2 \times 10^8\times$

Default rules of thumb

$d_\varepsilon \leq 4$: Gauss–Hermite, $Q = 5$. $5 \leq d_\varepsilon \lesssim 30$: Stroud-3. Beyond that, or fat tails: quasi-Monte Carlo.

Why this is cheap to get right

Only the quadrature **kernel** changes: same loss, same training loop. Validate by tightening the rule and checking that the Euler-error histogram does not move. Degree-3 exactness usually suffices near 10^{-3} on IRBC-class problems (Pichler 2011; Maliar & Maliar 2014), but that is empirical.

Implementation Gotchas That Cost a Day Each

Detach the simulated states

The path is generated *using* $\mathcal{N}_\theta$. If the states are not detached from the graph, autodiff differentiates through the simulation itself. Treat simulated states as fixed data, as in Azinovic et al. (2022).

Decide about $t+1$ deliberately

Whether the network at $t+1$ carries gradient or is frozen is the target-network question from fitted value iteration. Either is defensible; silently switching between them is not.

The rest of the checklist

- ▶ **Rescale the quadrature nodes** for the probabilists' convention.
- ▶ **Standardize the inputs** ($K_t, \ln z_t$) before the first layer; capital and log productivity live on very different scales.
- ▶ **Burn in** the simulated path before sampling from it, so training points come from the ergodic distribution and not from x_0 .
- ▶ **Evaluate the same network at every quadrature node**: Q extra forward passes per state, which is the dominant cost per step.

Diagnostics: How Do We Know Training Worked?

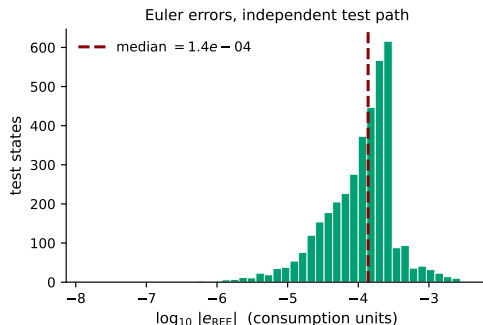
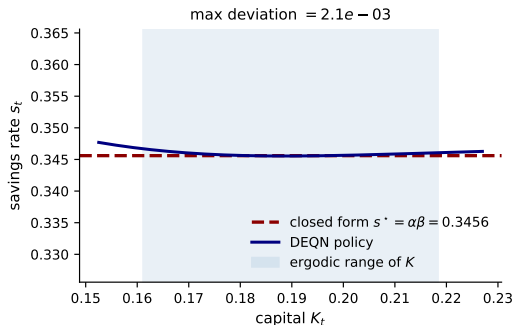
Never trust a loss curve alone. On an independent *test* simulation:

1. **Euler-error distribution.** Histogram $\log_{10} |e_{\text{REE}}|$ on train and test states, in consumption units; both should peak near 10^{-3} or below, with no fatter right tail on test.
2. **Policy-function benchmark.** Overlay the DEQN policy on the analytic $s^* = \alpha\beta$, or on a trusted reference from time iteration or perturbation.
3. **Ergodic-set coverage.** Scatter test states in $(K_t, \ln z_t)$: the cloud should trace the training ergodic set, neither collapsing to a point nor spreading out like a grid.
4. **Accounting and moments.** Verify budget, market clearing and non-negativity. For heterogeneous-agent models, report **den Haan (2010)** statistics.

There is no convergence theorem

Time iteration is a contraction; DEQN training is not. The sampling distribution moves as the policy moves, so “the loss stopped falling” proves nothing. Stop on the *test* Euler-error quantiles, and report them.

Validating a DEQN Against the Closed Form



Own computation. Stochastic Brock–Mirman, $\alpha = 0.36$, $\beta = 0.96$, $\varrho = 0.8$, $\sigma_z = 0.02$, so $s^* = 0.3456$; a 2×32 tanh network with a sigmoid head, 5-node Gauss–Hermite quadrature, expectation-inside loss. **Left:** learned savings rate against the closed form. **Right:** $\log_{10} |e_{REE}|$ on an independent test path. Note that the policy drifts *outside* the ergodic range, where it was never trained: a preview of Part VI.

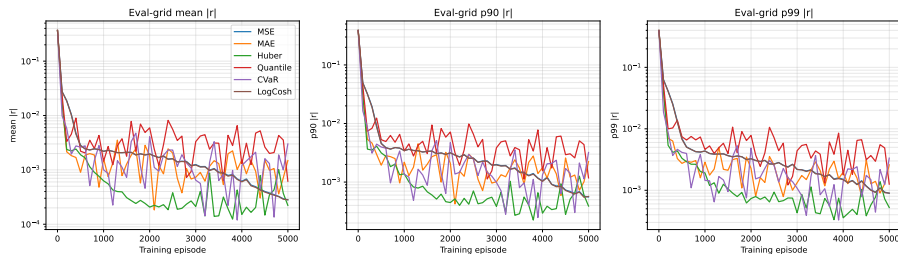
Choice of Training Loss: Six Kernels

The loss kernel is a **modeling decision**, not a default. Write $r = \mathcal{R}(\mathbf{x})$ for the scalar residual at a state; same residual, same network, same data, only the reduction $r \mapsto \mathcal{L}$ changes:

Kernel	Formula	Character
MSE	$\frac{1}{B} \sum r^2$	quadratic, canonical default
MAE	$\frac{1}{B} \sum r $	constant gradient, stalls at small r
Huber	quadratic for $ r \leq \delta_H$, linear above	robust hybrid, but kink at $r = \delta_H$
Quantile	pinball at $\tau = 0.9$	trains the upper τ -quantile only
CVaR	mean of the top 10% of $ r $	explicitly trains the worst-case states
LogCosh	$\frac{1}{B} \sum \log \cosh(r)$	C^∞ : $\frac{1}{2}r^2$ near 0, $ r - \log 2$ in the tails

Same Brock–Mirman model with full depreciation, so $s^* = \alpha\beta$ is in closed form; same network (2×32 Swish, sigmoid head); same Adam optimizer; same training stream. Six runs differ *only* in the reduction applied to the residual vector.

Loss Kernels: Convergence on Brock–Mirman



Own computation. Mean, p_{90} and p_{99} of the absolute relative Euler error on a fixed evaluation grid against training episode, log scale, 5,000 episodes per kernel.

- **LogCosh and MSE** reach the lowest mean residual here, **Huber and CVaR** the narrowest tail, and **MAE stalls** because its gradient magnitude does not shrink with the residual.
- This is one model with one architecture and the values are seed-sensitive. MSE stays the default; LogCosh is worth benchmarking against it; use Huber, Quantile or CVaR when the worst case matters.
- **Rank on the economic metric:** not the training loss: the two rankings need not agree.

Huge Potential, but Not Yet Dynare

- ▶ **The promise:** grid-free, high-dimensional, kinks and all, as the benchmark confirms.
- ▶ **The catch:** not yet a turnkey solver, and seed-sensitive.

Five pathologies of self-supervised training:

- Loss-scale imbalance:** normalize and adapt the weights. (Part IV)
- Architecture choice:** search the candidate space. (Part V)
- Lazy surrogate learning:** anchors and distillation. (Part VI)
- Learning from scratch:** freeze a guess, learn the correction. (Part VI)
- Self-confirming coverage:** audit on a coverage measure. (Part VI)

Seed dependence

Same model, same code,
different seed: one run
converges and verifies, another
stalls. Reproducibility is part of
the method.

Part VI

Further Pathologies, HPC, and Wrap-Up

Three Pathologies Left

Where we are:

- (a) Loss-scale imbalance **done, Part IV**
- (b) Architecture choice **done, Part V**
- (c) **Lazy surrogate learning**
- (d) **Learning from scratch**
- (e) **Self-confirming coverage**

The last three share a feature: the loss looks fine while the policy is wrong. None of them is visible on a training curve.

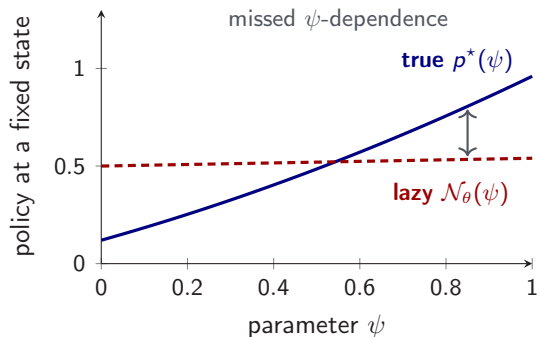
The common thread

Residual-only training measures the policy *where the policy itself decides to look*. Every remedy in this part changes *what is measured*, not the economics.

After the pathologies: data parallelism, the hands-on preview, and the wrap-up.

Lazy Learning: Low Residual, Wrong Policy

- **Symptom.** Lazy training is a typical pitfall of **self-supervised** (residual-only) learning: it drives the equilibrium residual down while the policy stays inaccurate across the state and parameter domain.
- Fix a state and vary a *parameter* ψ : a taste, a tax, or a shock size.
- The true policy moves with ψ ; the lazy surrogate stays *flat*, so it gets the level right on average and the **sensitivity** wrong.



At a fixed state the true policy varies with ψ (blue), while the lazy surrogate is almost flat (red dashed). The loss is low and the parameter dependence is weak: the mask is convincing only on average.

The Fix: Anchors and Distillation

Fix. Augment the self-supervised residual with a small **semi-supervised distillation** term on a few trusted anchor states:

$$J^{\text{dist}}(\theta) = J(\theta) + \lambda_a \sum_{\mathbf{x}_a \in \mathcal{A}} \|\mathcal{N}_\theta(\mathbf{x}_a) - p^*(\mathbf{x}_a)\|^2.$$

- ▶ $\mathcal{A}$: a handful of anchor states.
- ▶ p^* : reference policies from an accurate solver, such as perturbation, a low-dimensional solve, or a converged reference run.
- ▶ λ_a is one more weight, so Part IV applies to it as well.

Remedy

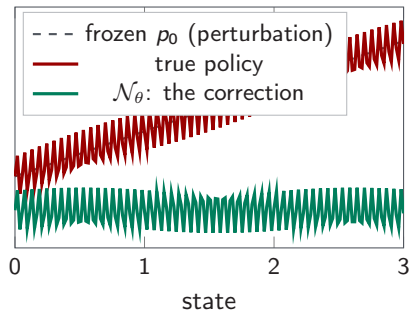
In the deep uncertainty-quantification application of Friedl et al. (2023), about **10 to 20 anchors** are typically enough to stabilize training, and leave-one-out diagnostics show the distilled surrogate agreeing with the reference policies to **within one percent**.

Residual Learning: Freeze and Correct

Idea. Do not ask the network for the *whole* solution, only for the **correction** to a cheap baseline:

$$p(\mathbf{x}) = \underbrace{p_0(\mathbf{x})}_{\text{frozen guess}} + \mathcal{N}_\theta(\mathbf{x}).$$

- ▶ p_0 : a cheap guess (steady state, perturbation, certainty equivalent).
- ▶ The network learns only the *small, nonlinear* part, which is easier than the level and the shape at once.
- ▶ The solution is sane from iteration zero, which matters when a bad start diverges.
- ▶ → **The network is a proofreader, not an author.**



The network only fits the small wiggle: frozen guess plus learned correction.

Residual Training Only Tests States the Policy Visits

Every DEQN solver is **self-supervised**: it drives an equation residual to zero on the states its *own* policy generates,

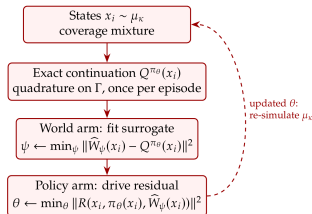
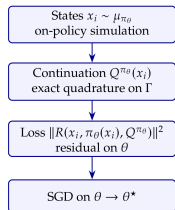
$$J(\theta) = \mathbb{E}_{\mathbf{x} \sim \mu_\theta} \|\mathcal{R}(\mathbf{x}, \mathcal{N}_\theta(\mathbf{x}))\|^2,$$

so the sampling measure μ_θ is generated **by the candidate policy itself**.

- ▶ The residual is only ever tested where the current policy already goes.
- ▶ A low loss says **nothing** about the states the policy avoids.
- ▶ A wrong policy visits the wrong states, so the error **hides exactly where it is never measured**.
- ▶ The loss **confirms itself**: a property of residual-only training, shared by every method in this lecture, and not a DEQN-specific flaw.

Equilibrium World Models: A Disciplined Dreamer

Equilibrium World Models (EWM) keep the economics, changing **where** the residual is imposed.



Left: an ordinary residual solver, measuring along its own path. Right: EWM, measuring on a broader coverage measure. Schaab & Scheidegger (2026), work in progress; Dreamer lineage: Ha & Schmidhuber (2018), Hafner et al. (2023).

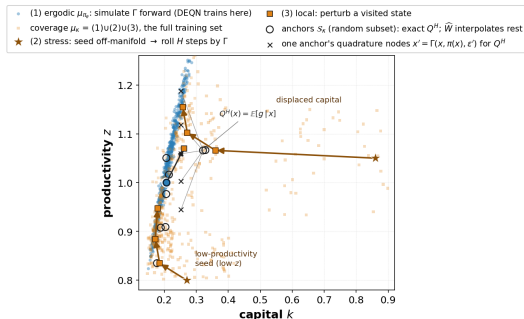
New: a coverage measure

$\mu_{\text{cov}} = c_1 \mu_\theta + c_2 \mu^{\text{stress}} + c_3 \mu^{\text{local}}$ and a learned continuation value, so the residual can be evaluated off-path.

Unchanged: the law of motion and the residual.

The audit is stronger because it runs on μ_{cov} , not because the economics changed.

What the Audit Buys: Off-Path Credibility



Coverage in (K, z) : the ergodic path plus stressed and counterfactual states, each from the exact law of motion. Schaab & Scheidegger (2026).

The economics is tested in **rare, stressed and counterfactual** regions, exactly where pathwise training is weakest.

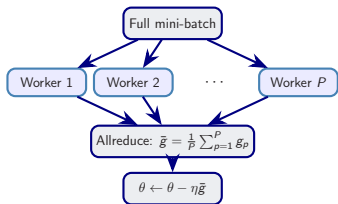
- In our experiments on **rare-disaster** models, the standard solver fails to find a verified solution across essentially every seed.
- With coverage, EWM solves in most runs and **audits** the error on the disaster region.

Not a new equilibrium concept, but a stronger certificate for the same one.

Data Parallelism for DEQNs

“To pull a bigger wagon, add more oxen rather than grow a gigantic ox.”

a folk aphorism of parallel computing, usually attributed to Seymour Cray



Idea: split the mini-batch across **multiple workers**. Each computes a local gradient on its slice, and an **allreduce** averages them before the update.

- ▶ Implemented with Horovod or MPI.
- ▶ Communication costs $\mathcal{O}(|\theta|)$ per step, **independent of the data size**: the network is exchanged, not the batch.
- ▶ Synchronous data parallelism is close to linear for moderate worker counts; large clusters become communication-limited.

Summary: What to Take Away

The method:

1. DEQNs replace **labels** with equations, **grids** with a network, and **time iteration** with SGD on simulated paths.
2. Enforce what you can *exactly* in the architecture; take the expectation *inside* the residual, and only then square it.
3. Validate against a benchmark; diagnose with Euler errors, never the loss curve.

The practice:

4. **Loss-scale imbalance** is the most common silent failure: normalize and rebalance.
5. **Architecture is a numerical choice**: search it, budget it, report it.
6. A low residual can still be a **lazy**, self-confirming policy: anchor it, audit off-path.

The one-sentence version. Solving an economic model becomes **training a neural network**, and then all the engineering that makes that training reliable becomes part of the numerical method.

Hands-On: Build a DEQN Yourself

You will build and train DEQNs yourself, starting from Brock–Mirman:

1. **Deterministic Brock–Mirman:** exogenous uniform sampling of states; compare the trained policy against the closed form $s^* = \alpha\beta$, and reproduce the validation figure from Part III. 02_01
2. **Stochastic Brock–Mirman** ($\varrho > 0$, $\sigma_z > 0$): **simulation-based sampling** on the ergodic set, with Gauss–Hermite quadrature for $\mathbb{E}_t[\cdot]$, including the node rescaling. 02_02
3. **Extensions:** endogenous labor with a non-negativity constraint, which introduces a *Karush–Kuhn–Tucker* multiplier $\nu_t \geq 0$ and a second loss component. 02_03, 02_04

Loss kernels are compared in 02_05. Architecture search and loss balancing follow in Session 3 (03_01, 03_02, 03_03), on a Genz testbed and then on this Brock–Mirman model.

Materials: [course GitHub repository](#) · **Platform:** [nuvolos.cloud](#) (no local setup required)

Exercise 3 in One Slide: KKT Without a Kink

The problem. Complementary slackness bundles three conditions at once,

$$\nu \geq 0, \quad g \geq 0, \quad \nu g = 0,$$

and inequalities are awkward inside a smooth loss.

The trick. The *Fischer–Burmeister* function (Fischer, 1992) packages all three into one residual:

$$\varphi_{\text{FB}}(a, b) = a + b - \sqrt{a^2 + b^2}, \quad \varphi_{\text{FB}}(a, b) = 0 \iff a, b \geq 0, \quad ab = 0.$$

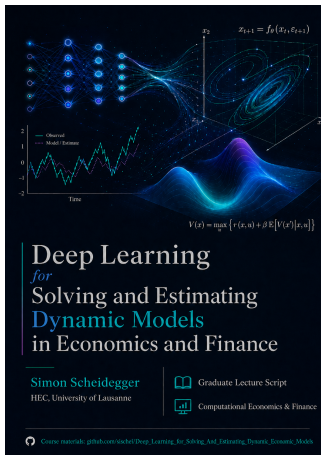
Why the equivalence holds

For $a, b \geq 0$, $a + b = \sqrt{a^2 + b^2}$ if and only if $(a + b)^2 = a^2 + b^2$, that is $2ab = 0$. The function is differentiable everywhere except at the origin, and autodiff handles it directly.

Where Part IV comes back

The exercise loss adds a squared *Euler* residual to a squared *Fischer–Burmeister* residual. The two live at different scales, which makes it a textbook case for the **loss balancing** of Part IV.

Want More Details? The Full Book



Deep Learning for Solving and Estimating Dynamic Models in Economics and Finance

A book and graduate lecture script that develops these methods in full.

- ▶ **18 lectures:** from foundations to research-level applications.
- ▶ **Everything is in code:** each lecture ships with runnable notebooks.
- ▶ Covers what this lecture could only sketch: heterogeneous agents, occasionally binding constraints, estimation, and uncertainty quantification.

Freely available: [the book repository on GitHub](#)

Questions?



Simon Scheidegger

HEC, University of Lausanne

<https://sischei.github.io>

Next up: architecture search and loss balancing
(Session 3)

Materials:

github.com/sischei/deep_learning_padova_0926

Platform: nuvolos.cloud

Backup slides

GradNorm: Balance the Gradients, Not the Losses

Idea (Chen et al., 2018): adjust ω_k so that the *gradient norms* are balanced,

$$\|\nabla_{\mathbf{W}} \omega_k \mathcal{L}_k\|_2 \approx \bar{G} \cdot \kappa_k^{(m)},$$

by descending on the auxiliary loss

$$\mathcal{L}_{\text{grad}} = \sum_{k=1}^M \left| \|\nabla_{\mathbf{W}} \omega_k \mathcal{L}_k\|_2 - \bar{G} \cdot \kappa_k^{(m)} \right|,$$

with the target $\bar{G} \cdot \kappa_k^{(m)}$ treated as a **constant** when differentiating. The weights are learnable, $\omega_k \leftarrow \omega_k - \eta_{\omega} \nabla_{\omega_k} \mathcal{L}_{\text{grad}}$, then renormalized to $\sum_k \omega_k = M$.

where

- ▶ $\mathbf{W}$ is a **chosen subset of shared weights**, typically the last shared layer, which is what makes the method affordable;
- ▶ $\bar{G}$ is the average gradient norm over components;
- ▶ $\kappa_k^{(m)} = (\tilde{\mathcal{L}}_k^{(m)} / \tilde{\mathcal{L}}^{(m)})^{\alpha_{\text{gn}}}$ is the relative inverse training rate, with $\tilde{\mathcal{L}}_k^{(m)} = \mathcal{L}_k^{(m)} / \mathcal{L}_k^{(0)}$ and $\alpha_{\text{gn}} \approx 1.5$.

Pros: principled, and it addresses gradient competition directly.

Cons: needs per-component gradients, so it is more expensive to run and more intricate to implement.

Bayesian Optimization

Idea: build a *surrogate* from configuration to loss, and use it to pick the next trial.

1. Fit a Gaussian process to the observed trials
2. Compute an **acquisition function**, such as Expected Improvement
3. Evaluate the best configuration, update, repeat

Advantage: **sample-efficient**: it exploits structure in the response surface.

Expected Improvement

$$\text{EI}(\mathbf{x}) = \mathbb{E}[\max(f^* - f(\mathbf{x}), 0)]$$

with f^* the best observed value: it balances *exploration* against *exploitation*.

Cons: inherently sequential, so harder to parallelize; the surrogate fit costs $\mathcal{O}(n^3)$ in the number of trials and degrades in high dimensions.

Search Strategies Compared

Property	Random	Bayesian	Hyperband
Trials to a given quality	Baseline	Fewest	Few
Cost per trial	Full training	Full training + surrogate	Early stopping
Parallelization	Fully parallel	Sequential	Mostly parallel

- ▶ Comparing raw cost in the budget n is meaningless, since n is chosen. The meaningful axis is how many trials a method needs to reach a given quality.
- ▶ Grid search is omitted: it is dominated by random search on every axis here.
- ▶ Hyperband samples configurations at random, so it handles interactions exactly as well as random search. What it adds is early stopping, not a better proposal.

Production tooling wraps all of these: Optuna, Ray Tune, KerasTuner, Ax.